HKUST MSc(AIE) Program Course Syllabus Template

Course Code: MAIE 5103
Course Title: Artificial Intelligence Ethics
No. of Credits: 3

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Course Description

This course delves into the ethical considerations involved in designing, developing, and deploying artificial intelligence (AI) systems. Students will investigate the societal impacts of AI and the ethical dilemmas that emerge in areas such as privacy, bias, transparency, and accountability. Through case studies and discussions, students will build a framework for ethical decision-making in the development and implementation of AI systems.

Course Intended Learning Outcomes (CILOs)

On successful completion of this course, students will be able to:

1. Acquire a comprehensive understanding of AI ethical considerations and social implications;
2. Develop own perspectives on the foundational ethical questions surrounding emerging technologies;
3. Critically analyze impacts and societal challenges of AI;
4. Apply knowledge to real-world AI-related issues and propose solutions;
5. Enhance written and oral communication skills to effectively discuss ideas related to AI ethics.

Assessments

Assessment Task	Assessment Weighting (%)	Due date *(MM/DD/YYYY)*	Mapped ILOs *(Write CILO-1, CILO-2, etc.)*
Participation	20	To be held in class	CILO-3, CIO-4, CILO-5
Quizzes (4 quizzes in total)	10	To be held in class	CILO-1, CILO-2, CILO-3
AI news presentation (Individual)	10	Around end-October	CILO-2, CILO-3, CILO-4
Reading response (Individual)	15	Around mid-October	CILO-1, CILO-2, CILO-5
Case study analysis (Individual)	15	Around mid-November	CILO-3, CILO-4, CILO-5
Final project report (Group)	15	Around early-December	CILO-1, CILO-2, CILO-3, CILO-4, CLIO-5

(Last Updated on May 8, 2025)

Final project presentation (Group)	15	To be held in class	CILO-1, CILO-2, CILO-3, CILO-4, CLIO-5

Week-by-week Content

Week 1: Course Overview & Lecture 1: Introduction to AI Ethics
Week 2: Lecture 2: Privacy and Data Protection
Week 3: Lecture 3: Accountability, Responsibility, Transparency
Week 4: Lecture 4: Fairness and Bias (Part 1) & Tutorial on Academic Writing
Week 5: Lecture 4: Fairness and Bias (Part 2)
Week 6: Lecture 5: Beyond Bias (Part 1) & Tutorial on News Presentation
Week 7: Lecture 5: Beyond Bias (Part 2)
Week 8: Lecture 6: The Ethics of Automation & Lecture 7: Safety and Security
Week 9: Lecture 8: AI in Social Media
Week 10: Lecture 9: AI in Military & Lecture 10: AI in Politics
Week 11: Lecture 11: AI Policy and Regulations
Week 12: Final Project Presentations
Week 13: Final Project Presentations
(Order and contents may be adjusted later according to teaching progress.)

Academic Integrity

Students are expected to adhere to the university's academic integrity policy. Students are expected to uphold HKUST's Academic Honor Code and to maintain the highest standards of academic integrity. The University has zero tolerance of academic misconduct. Please refer to Academic Integrity | HKUST – Academic Registry for the University's definition of plagiarism and ways to avoid cheating and plagiarism.

Required Texts and Materials

Reference books (all available in HKUST library online resources or book collections):

- Virginia Dignum. Responsible Artificial Intelligence: How to Develop and Use AI in a Responsible Way. Springer, 2019.
- Mark Cockelbergh. AI Ethics. MIT Press, 2020.
- Edited by Markus D. Dubber, Frank Pasquale, and Sunit Das. The Oxford Handbook of Ethics of AI. Oxford University Press, 2020.
- Virginia Eubanks. Automating Inequality: How High-tech Tools Profile, Police, and Punish the Poor. St. Martin's Press, 2018.
- Cathy O'Neil. Weapons of Math Destruction: How Big Data Increases Inequality and Threatens Democracy. Crown, 2016.

Every week, a list of readings will be provided and released in canvas as additional resources.

Course AI Policy

Students are not allowed to use generative AI (such as ChatGPT) to directly produce any content of assignments. This will be treated as plagiarism. Students might use generative AI as they would use a human collaborator (e.g., to aid in searching for information). Any interactions with these tools must be credited in assignments.